Computer Vision-Based Automated Detection of Building Setback Violations in RAJUK Jurisdictions

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Abstract

Dhaka's rapid and largely unregulated urban growth has created a city where building setback violations are commonplace, quietly eroding pedestrian safety, blocking emergency routes, and degrading the quality of life in neighborhoods across the metropolis. The regulations set by the Rajdhani Unnayan Kartripakkha (RAJUK) exist precisely to prevent these harms, yet enforcing them through manual inspection is slow, inconsistent, and simply impossible to scale across a city of this size. In this paper, we describe a computer vision-based system designed to automatically detect setback conditions directly from photographs taken by field personnel on the ground. To support this work, we built a purpose-specific dataset of 372 geo-tagged street-level images, each labeled as *positive* (violation), *moderate* (borderline), or *negative* (compliant) based on RAJUK's prescribed setback norms, and each annotated with a measured building-to-boundary distance. We then trained and evaluated two deep learning models on this dataset: a fully-connected Multi-Layer Perceptron (MLP) trained from scratch on raw pixels as a lower-bound baseline, and a ResNet18 convolutional network fine-tuned from ImageNet weights. On a held-out test set of 56 real images, the MLP reached 50.0% accuracy (macro-F1 0.42). In an improved CNN pipeline that uses a larger, 70:20:10 subject-level split together with hard-example mining of previously misclassified validation images, the ResNet18 transfer-learning model reached **66.6%** test accuracy (macro-F1 0.53) on a much larger 1,255-image held-out test set, with a training/validation/test error profile of 0.00%/31.99%/33.39%, confirming the value of convolutional feature extraction for this task. Our early results show that image-based automated compliance monitoring is genuinely feasible, offering planning authorities a scalable and cost-effective tool to prioritize inspections

and strengthen enforcement, a meaningful step toward more liveable, better-managed urban development in one of the most densely populated cities on Earth.

Keywords: Computer Vision, Building Setback Violations, RAJUK, Convolutional Neural Networks, Transfer Learning, Image Classification, Urban Compliance Monitoring, Deep Learning, Dhaka.

1 Introduction

Dhaka is one of the fastest-growing megacities on the planet. With a population that has now surpassed 22 million, the city's built environment continues to expand at a pace that planning systems struggle to keep up with [1]. One consequence of this growth is the widespread flouting of urban regulations — and among these, violations of building setback rules are particularly visible and damaging. RAJUK, the principal authority responsible for guiding construction activity across greater Dhaka and its surrounding areas, has long sought ways to bring this problem under control [2].

A *building setback* is the minimum legally required gap between a structure and its plot boundary or the adjacent road. Under RAJUK's Dhaka Metropolitan Development Plan (DMDP) and the Bangladesh National Building Code (BNBC), the specific distances vary with road width, land use zone, and building height [3]. When these distances are not respected, buildings encroach onto public footpaths and rights-of-way. The results are predictable: hazardous conditions for pedestrians, blocked access for emergency vehicles, and a steady deterioration of the streetscape.

The scale of the problem, however, makes it very difficult to address. Today's primary enforcement tool, manual inspection by RAJUK field officers, is time-consuming, resource-intensive, and prone to inconsistency. What is needed is an automated approach that can survey the city's vast and varied built fabric quickly, objectively, and repeatedly.

Deep learning has already shown tremendous promise across domains that share these characteristics, from agriculture [6, 7] and road infrastructure [5] to urban analytics [4], often relying on transfer learning to work well with limited labeled data [8]. Yet no prior work has tackled building setback violation detection from street-level imagery within a South Asian regulatory context. Most studies on urban violations use aerial or satellite views [9], which do not capture the facade-to-road proximity that actually defines a setback violation

at ground level.

This paper addresses that gap. We present a neural network-based framework for identifying building setback conditions from photographs taken at street level, supported by a new labeled dataset of images collected across RAJUK's jurisdiction. By using smartphone photographs taken by field personnel rather than satellite data, we have aimed for an approach that is practical and deployable from the outset.

Our specific contributions are:

- **A new real-world dataset:** We conducted on-foot surveys across Dhaka's roads and neighborhoods, collecting and labeling 372 building facade images as positive (violation), moderate (borderline), or negative (compliant), each with a measured building-to-boundary setback distance.
- **A structured annotation protocol:** We developed a three-level labeling framework grounded in RAJUK and BNBC setback norms [2, 3], making annotations reproducible and directly tied to regulation.
- **A systematic model comparison:** We benchmark a from-scratch fully-connected MLP baseline against a ResNet18 [14] transfer-learning model under leakage-free protocols, establishing reference performance levels for this new task.
- **A practical deployment pathway:** We lay the groundwork for a mobile or web-based tool that would allow RAJUK field staff to receive immediate violation predictions from images they capture on site.

2 Related Work

A growing body of research applies deep learning to urban compliance monitoring, and several strands are directly relevant to our work.

Footpath and sidewalk encroachment detection. The most closely related work is that of Lubaina et al. [4], who introduced *FootpathVision* the first publicly available labeled dataset for sidewalk encroachment detection in Dhaka. Containing 1,238 images annotated with COCO-style bounding boxes, the dataset supports both binary (encroached vs. clear) and multi-class (illegal stalls, illegal parking, clear) tasks. Their evaluation of CNN architectures including MobileNetV2, EfficientNetB0, ResNet50, VGG16, and DenseNet121 showed that binary classification is robust and consistent, while multi-class detection runs into class imbalance difficulties. This work directly demon-

strates that locally collected, street-level datasets can support effective urban compliance monitoring in Dhaka, a key precedent for our own approach.

Building and construction violation detection. Most existing work on detecting illegal buildings relies on satellite or aerial imagery. Huang et al. [9] used multi-temporal satellite images to identify unauthorized construction through change detection pipelines. Varghese et al. [10] proposed ChangeNet, a deep learning architecture for visual change detection in public urban spaces. While these aerial approaches are powerful in their own right, they fundamentally cannot capture the facade-to-road proximity that defines a setback violation, which is precisely why street-level imagery is essential for our task.

Road and infrastructure damage detection. Waki et al. [5] built a YOLOv9-based pipeline for road damage and manhole detection in Dhaka, using polygonal annotations on a dataset of over 1,000 images and achieving strong results (86.7% F1-score for broken road detection). Their work confirms the practical value of locally collected Bangladeshi datasets for automated infrastructure monitoring. In a similar vein, Maeda et al. [11] showed that smartphone-captured road damage images can support effective deep learning-based detection, validating the mobile-captured data collection strategy we adopt here.

Sidewalk accessibility and pedestrian infrastructure. The Project Sidewalk dataset [12], with more than 300,000 image-based labels, showed that modified ResNet architectures can approach human-level performance in identifying accessibility issues from street-view panoramas. Hosseini et al. [13] extended this direction by combining crowd-sourced data with AI-assisted sidewalk mapping. Together, these works underscore the wider potential of vision-based systems for monitoring pedestrian infrastructure.

Transfer learning on small datasets. With large-scale annotation being both expensive and time-consuming, fine-tuning ImageNet-pretrained models has become the standard approach for domain-specific vision tasks with limited labeled data. The residual connections in ResNet [14] make it particularly well-suited to fine-tuning. MobileNetV2 [15] offers a lightweight alternative designed for mobile and edge deployment. EfficientNet [16] goes further, jointly scaling network depth, width, and resolution for better efficiency. Hasan et al. [8] demonstrated that genetic algorithm-guided selection of update blocks can further improve transfer learning outcomes for domain-specific tasks. We adopt ResNet18 with partial fine-tuning as the convolutional counterpart to our pixel-based baseline.

What is missing. Despite this breadth of related work, no study has specif-

ically targeted building setback violation detection from street-level imagery within a South Asian regulatory framework. Our work addresses this gap directly: we focus on setback violations as legally defined by RAJUK [2], use photographs practical for field deployment, and build a dataset grounded in Dhaka's actual urban morphology.

3 Dataset and Annotation

To support our experiments, we built a dataset we call **RAJUKSetbackBD**. It consists of 372 labeled images collected by team members through on-foot surveys of selected neighborhoods within RAJUK's jurisdiction, all captured using mobile phone cameras.

3.1 Study Area and Regulatory Context

Our study covers buildings within RAJUK's jurisdiction, which spans the Dhaka Metropolitan Area (DMA). The relevant regulations codified in the Dhaka Metropolitan Building Construction Rules (2008) [2] and the BNBC (2020) [3] specify minimum setback distances from road boundaries, with requirements varying according to road width and land use zone. Under this framework, a building is considered a setback violator if its ground-floor structure, or any permanent extension such as a boundary wall, ramp, or covered veranda, encroaches on or beyond the legally mandated setback line.

3.2 Image Acquisition

All images were captured using smartphone cameras at a minimum resolution of 1920×1080 pixels, held in landscape orientation, with the photographer standing on the opposite side of the road to ensure full facade visibility. We made no attempt to control real-world conditions: images were taken at different times of day (353 daytime and 19 nighttime captures), under a range of lighting conditions (sunny, overcast, shaded), and with typical street obstructions such as parked vehicles, vendors, and pedestrians present in frame. This deliberate variability mirrors the conditions that a deployed system would actually encounter, and follows the smartphone-based data collection strategy established in earlier work on urban infrastructure monitoring [11].

For each image we recorded the neighborhood or area, the time of capture (day or night), and two measured distances: the building-to-boundary setback distance and the camera-to-facade distance, both in feet. The recorded setback distances range from 0 to 12 feet, providing a quantitative anchor for the categorical compliance label assigned by the annotators.

3.3 Annotation Protocol

The RAJUKSetbackBD dataset contains 372 smartphone-captured images drawn from selected neighborhoods in the Dhaka Metropolitan Area, collected under both daytime and nighttime conditions to capture the full range of visibility and lighting environments that a real-world system would face. The minimum image resolution is 1920×1080 pixels.

Rather than using bounding box annotations (as in, for example, [4, 5]), we adopted a structured image-level labeling protocol grounded directly in RAJUK's setback regulations [2, 3]. Each image was assigned one of three labels:

- **Positive / VIOLATION:** The building's ground-floor structure, or a permanent extension such as a boundary wall, ramp, or covered veranda, is visually judged to be encroaching on or beyond the mandated setback space between the facade and the road or footpath boundary. The dataset contains 61 images in this class.
- **Negative / COMPLIANT:** A clear and unobstructed setback space is visibly present between the building facade and the road or footpath boundary, indicating apparent consistency with RAJUK's prescribed norms [2]. The dataset contains 229 images in this class.
- **Moderate / Borderline:** The image presents an ambiguous or borderline condition where setback compliance cannot be determined with confidence from visual inspection alone. This may result from partial occlusion, a restricted viewing angle, poor lighting, roadside obstructions, or genuine uncertainty about the exact plot boundary. The dataset contains 82 images in this class.

To summarize the CSV annotation format: **Positive** = **VIOLATION**, **Moderate** = **Borderline**, and **Negative** = **COMPLIANT**. The dataset spans sixteen neighborhoods across Dhaka, with the largest contributions from Khilgaon, Mirpur 10, Mohammadpur, Monsurabad, Changkharpul, and Dhanmondi, along with Wari, Mirpur 1, Mirpur 12, Bashundhara, Lalbag, Purana Paltan, Shyamoli, Segunbagicha, Puran Dhaka, and Bosila. All images were collected in 2026.

Each image was reviewed by members of our team against the visual evidence of setback space between the building facade and the road or footpath boundary, cross-referenced with the recorded setback distance. Where reviewers disagreed, or where the image did not provide enough visual clarity for a confident decision, the image was assigned to the Moderate class.

Figure 1 shows representative examples from each of the three annotation categories, illustrating how smartphone images can support visual setback assessment under varied urban and lighting conditions.



Figure 1: Representative labeled images from the RAJUKSetbackBD dataset, one row per class. Negative examples show buildings with a visible, unobstructed setback; Moderate examples show borderline or visually ambiguous cases; and Positive examples show apparent setback violations.

3.4 Dataset Statistics

Table 1 summarizes the composition of the collected dataset. The class distribution shows a marked imbalance toward the compliant (negative) class, which is consistent with what we observed in the field and which we address explicitly through weighted loss during training (Section 4).

Table 1: Summary statistics of the RAJUKSetbackBD dataset.

Attribute	**Value**
Total labeled images	372
VIOLATION (positive)	61
COMPLIANT (negative)	229
Moderate / borderline	82
Augmented images generated	5,968
Minimum resolution	1920×1080 px
Capture device	Smartphone cameras
Recorded metadata	Building-to-boundary distance, camera distance (feet)
Annotation format	Positive = VIOLATION; Moderate = borderline; Negative = COMPLIANT (CSV)
Study area	RAJUK jurisdiction, Dhaka Metropolitan Area
Neighborhoods covered	Khilgaon, Mirpur 10/1/12, Mohammadpur, Monsurabad, Changkharpul, Dhanmondi, Wari, Bashundhara, Lalbag, Purana Paltan, Shyamoli, Segunbagicha, Puran Dhaka, Bosila
Collection time	Day (353), Night (19)
Collection period	2026

Annotation was carried out by members of our research team. Each image was reviewed carefully against RAJUK's setback specifications [2, 3] and the recorded building-to-boundary distance to keep labeling errors to a minimum. The cases where reviewers disagreed were mostly those involving partial occlusion, poor lighting, or genuinely ambiguous facade-to-road proximity, and were resolved into the Moderate class.

The dataset is publicly released to support further research at [19]:

https://docs.google.com/spreadsheets/d/1_QZuP0QnBCnsbc-FAWuSNSJL57rJ3GxVhxL3k6gdM8c

4 Learning Algorithms

This section describes the two learning algorithms we implemented for classifying building setback conditions from images. Both models assign each input image to one of three categories: *negative* (compliant setback), *moderate* (borderline or ambiguous), and *positive* (apparent setback violation), and share the same underlying dataset, weighted cross-entropy loss, AdamW optimization, cosine-annealing learning-rate schedule, and early stopping, enabling fair comparison. The MLP baseline was trained on an NVIDIA T4 GPU (Google Colab) with a fixed random seed of 42; the improved CNN pipeline described in Section 4.3 was trained locally on an NVIDIA GPU using its own, independently constructed leakage-free split (Section 4.3.1).

4.1 Dataset and Leakage-Free Split

Labels are loaded from a structured annotation file [19]. A header-normalization step strips whitespace and non-alphanumeric characters from column names so the label column can be located reliably, and short codes (`P`, `M`, `N`) are expanded to the canonical class strings *positive*, *moderate*, and *negative*, encoded as integers 0, 1, 2. Of 372 annotated rows, all 372 yielded valid labels.

Images are drawn from two sources: 373 original field-survey photographs (372 of which carry a matching label) and 5,968 synthetically augmented copies generated offline to increase training-set diversity, a standard practice when labeled data is scarce [4, 8]. To prevent evaluation inflation, we split at the level of unique *subject* identifiers before any augmented image is assigned. Concretely, the real-image indices are partitioned into training, validation, and test subsets using stratified sampling; augmented copies are then assigned *exclusively* to the training subset, and augmented copies of any validation or test subject are discarded. This guarantees that no near-duplicate of a held-out

image is ever seen during training. The resulting partition sizes for the MLP baseline are shown in Table 2. (The improved CNN pipeline uses a revised, independent split protocol, described separately in Section 4.3.1.)

Table 2: Dataset partition sizes after leakage-free subject-level splitting (MLP baseline).

Split	Sources Included	Samples	Fraction
Training	Real + Augmented	4,420	97.5%
Validation	Real only	57	1.3%
Test	Real only	56	1.2%

To compensate for the majority-class imbalance, per-class loss weights are computed as the normalized inverse of class frequency in the training split,

$$w_c = \frac{1/n_c}{\sum_{c'} 1/n_{c'}},$$

where n_c is the number of training samples in class c. On the training split (negative: 2,720, moderate: 969, positive: 731), this yields weights of 0.1328 (negative), 0.3729 (moderate), and 0.4943 (positive), reflecting the relative scarcity of violation and borderline images.

4.2 Algorithm 1: Multi-Layer Perceptron (MLP)

4.2.1 Architecture

The MLP is a fully-connected baseline with no convolutional or attention components, designed to quantify how much can be achieved from raw pixel statistics alone, without the spatial inductive bias that networks like VGG16 [17], ResNet [14], or DenseNet [18] provide. Each image is resized to 128×128 pixels and flattened into a vector of length

$$d = 128 \times 128 \times 3 = 49{,}152.$$

Four hidden blocks of decreasing width each apply a linear transformation, one-dimensional batch normalization, ReLU activation, and dropout (rate = 0.40), followed by a single linear output layer:

$$49{,}152 \;\to\; 2{,}048 \;\to\; 1{,}024 \;\to\; 512 \;\to\; 256 \;\to\; 3.$$

The final layer produces three raw logits with no activation; the predicted class is the argmax of the logit vector. The network totals 103,428,099 trainable parameters.

4.2.2 Training and Results

Stochastic pixel-level augmentations (horizontal flip p=0.5, vertical flip p=0.3, and brightness, contrast, and saturation jitter) are applied on-the-fly during training; we deliberately avoid spatial augmentations such as rotation or cropping, since a flat MLP has no conception of spatial structure. Weighted cross-entropy loss compensates for class imbalance, and parameters are optimized with AdamW under a cosine-annealing schedule (LR $10^{-3} \rightarrow 10^{-6}$). A complete summary of hyperparameters is given in Table 3.

Stratified 5-fold cross-validation is run *exclusively on the 260 real training images* (208 train / 52 validation per fold), so that no augmented near-duplicate of a fold's validation sample can leak into its training partition. The per-fold best validation accuracies were 0.6154, 0.5577, 0.5192, 0.6154, and 0.6154, giving a mean of 0.5846 ± 0.0396. The best cross-validation checkpoint is then fine-tuned on the full 4,420-sample training split for five epochs at LR $= 2 \times 10^{-4}$. Training accuracy climbed steadily across the five fine-tuning epochs (45.2%, 64.8%, 79.7%, 86.5%, 89.3%), but final validation accuracy was only 42.1% and test accuracy 50.0%, indicating substantial overfitting. Results are therefore interpreted together with the cross-validated estimate rather than from the fine-tuned training accuracy alone.

Table 3: Training configuration — MLP baseline.

Parameter	**Value**
Input dimension	49,152 ($128 \times 128 \times 3$ flattened)
Hidden layer widths	2048, 1024, 512, 256
Activation / Norm	ReLU / Batch Norm (1-D)
Dropout rate	0.40
Total parameters	103,428,099
Loss function	Weighted Cross-Entropy
Optimiser	AdamW ($\lambda = 10^{-3}$)
LR schedule	Cosine Annealing ($10^{-3} \rightarrow 10^{-6}$)
Batch size	32
Max epochs / Patience	60 / 10
Cross-validation	Stratified 5-fold (real images only)
Fine-tuning	5 epochs at LR $= 2 \times 10^{-4}$
Hardware	NVIDIA T4 GPU

4.3 Algorithm 2: Convolutional Neural Network (CNN)

4.3.1 Revised Split Protocol and Hard-Example Mining

The improved CNN pipeline retrains ResNet18 with two methodological changes relative to the shared protocol in Section 4. First, rather than reserving only 2.5% of the data for evaluation, the subject-level split is rebalanced to an approximate 70:20:10 train:test:validation ratio (TEST_FRACTION = 0.20, VAL_FRACTION = 0.10, applied to the 356 uniquely labeled real subjects), giving much larger and more statistically reliable validation and test sets. As before, the split is performed on unique real-subject identifiers and each subject's augmented copies follow it into the same partition, so no augmented near-duplicate of a held-out subject ever appears in training. Of 6,341 total images (373 real, 17 of which lack a label match, plus 5,968 augmented), the initial leakage-free split allocates 4,420 images (69.7%) to training, 646 (10.2%) to validation, and 1,275 (20.1%) to test.

Second, we introduce a *hard-example mining* step: every validation image that a prior training run misclassified (457 rows recorded in `val_misclassified.csv`) is identified, and its real (raw) copy is moved into the training set unconditionally, while roughly 50% of that subject's misclassified augmented copies are also moved in (selected per subject to avoid duplicating the entire augmented set). This moves 27 real images and 220 augmented copies from validation, and a further 20 test images, into training, so the model is explicitly re-exposed to the visual conditions it previously struggled with. Table 4 shows the resulting split.

Table 4: Dataset partition sizes for the improved CNN pipeline, before and after hard-example mining.

Split	Before mining	After mining	Fraction (after)
Training	4,420	4,467	70.4%
Validation	646	619	9.8%
Test	1,275	1,255	19.8%

Class weights are recomputed on the post-mining training split (negative: 2,729, moderate: 997, positive: 741), giving weights of 0.1348 (negative), 0.3689 (moderate), and 0.4963 (positive), used in the weighted cross-entropy loss.

4.3.2 Architecture

The CNN is a ResNet18 [14] backbone initialized with ImageNet pretrained weights. With only ∼356 unique real subjects, training from scratch would overfit; transfer learning lets the network reuse low-level feature detectors (edges, textures, shapes) already learned from large-scale data. All layers except the final residual block (`layer4`) and the classification head are frozen, restricting gradient updates to the most task-relevant parameters — 8,395,267 of the 11,178,051 total parameters (75.1%) are trainable. The original fully-connected output layer is replaced by

$$512 \xrightarrow{\text{Dropout}(0.5)} \text{Linear}(512, 3),$$

yielding three output logits over the target classes. Images are resized to 224×224 pixels (ResNet18's native resolution) and normalized using ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$).

4.3.3 Training and Results

Taking advantage of the CNN's spatial feature extraction, we apply a richer set of on-the-fly augmentations during training: random horizontal and vertical flips, random rotation ($\pm 15°$), color jitter (brightness, contrast, saturation, hue), random resized crop, and Gaussian blur. Validation and test images pass through a deterministic resize-and-normalize pipeline only. Weighted cross-entropy loss, AdamW optimization, and cosine annealing (LR $3 \times 10^{-4} \rightarrow 10^{-6}$) are used throughout. A complete summary of hyperparameters is given in Table 6.

Cross-validation. Stratified 3-fold cross-validation is run over the post-mining training split (4,467 images; ≈2,978 train / 1,489 validation per fold). Because each real building appears alongside its augmented copies in this pool, the resulting cross-validation accuracy is very high — fold accuracies of 1.0000, 0.9993, and 1.0000, giving a mean of 0.9998 ± 0.0003 — but this figure is *optimistically biased*, since augmented variants of a building in a fold's training partition strongly resemble those in its validation partition. We therefore treat the held-out test accuracy on the independent test split as the honest measure of generalization.

Fine-tuning. The best cross-validation checkpoint is fine-tuned on the full post-mining training split for five further epochs at LR $= 6 \times 10^{-5}$ (= LR/5). Training accuracy during fine-tuning is already high and stable across all five epochs (99.51%, 99.17%, 99.57%, 99.49%, 99.46%), reflecting that the cross-validated backbone has already converged. Evaluated on the held-out valida-

tion and test sets, the fine-tuned model attains **68.01% validation accuracy** (loss 1.7800) and **66.61% test accuracy** (loss 1.9845).

Final train / validation / test error report. To report training error on a strictly comparable footing (i.e. measured with the same deterministic resize-and-normalize pipeline used for validation and test, rather than the augmented training-time view), we re-evaluate the fine-tuned model on the training set with augmentation disabled. Table 5 and Figure 5 summarize the resulting error rates.

Table 5: Final error report for the improved CNN (fine-tuned best cross-validation checkpoint), evaluated with a deterministic (non-augmented) transform on all three splits.

Split	Samples	Loss	Accuracy	Error rate
Training	4,467	0.0009	100.00%	0.00%
Validation	619	1.7800	68.01%	31.99%
Test	1,255	1.9845	66.61%	33.39%

The training error of 0.00% confirms that the fine-tuned network has fully memorized the training split (including its hard-mined examples), while the validation and test error rates of roughly 32–33% show the genuine, non-trivial generalization gap that remains on held-out subjects. Despite this gap, both the validation and test accuracies (68.0% and 66.6%) are noticeably higher than in the original 2.5% held-out protocol, reflecting the benefit of a larger, more statistically stable evaluation set together with hard-example mining.

4.4 Evaluation Protocol

The MLP is evaluated on its own 56-sample held-out test set of real images, as described in Section 4. The improved CNN pipeline is evaluated on its own, independently constructed 1,255-image held-out test set (Section 4.3.1), which is substantially larger and includes augmented copies of held-out subjects in addition to real photographs. Because the two evaluation sets differ in size and composition, absolute accuracy figures between the two models are not perfectly apples-to-apples; nonetheless, both protocols are leakage-free at the subject level, and both are reported so that the overall trend — convolutional features generalizing better than raw pixels — can still be assessed. For both models we report per-class precision, recall, and F1-score, and confusion matrices for validation and test sets. Learning curves are generated by retraining each model at several dataset-size checkpoints to assess data efficiency.

Table 6: Training configuration — CNN (ResNet18 transfer learning, improved pipeline).

Parameter	Value
Backbone	ResNet18 (ImageNet pretrained)
Frozen layers	All except `layer4` + head
Trainable parameters	8,395,267 (75.1% of 11,178,051)
Input size	$224 \times 224 \times 3$ (ImageNet normalization)
Classifier head	Dropout(0.5) + Linear(512, 3)
Loss function	Weighted Cross-Entropy
Optimiser	AdamW ($\lambda = 3 \times 10^{-4}$)
LR schedule	Cosine Annealing ($3 \times 10^{-4} \rightarrow 10^{-6}$)
Batch size	16
Max epochs / Patience	25 / 6
Weight decay	1×10^{-3}
Split ratio (train:test:val)	70% : 20% : 10% (subject-level)
Hard-example mining	Misclassified val/test subjects re-injected into train
Cross-validation	Stratified 3-fold
Fine-tuning	5 epochs at LR $= 6 \times 10^{-5}$
Hardware	NVIDIA GPU (local workstation)

5 Experimental Results

Table 7 compares the two models. The ResNet18 transfer-learning model outperforms the flat-pixel MLP baseline on every aggregate metric, improving test accuracy from 50.0% to 66.6% and macro-F1 from 0.42 to 0.53. The per-class breakdown in Table 8 shows that most of the gain comes from the negative (compliant) and positive (violation) classes; the moderate (borderline) class remains difficult for both models, which is expected given its inherent visual ambiguity.

Table 7: Performance comparison between the two models. The MLP is evaluated on its 56-image real-only test set; the improved CNN is evaluated on its own, larger 1,255-image test set (Section 4.3.1). The CNN cross-validation figure is optimistically biased because the folds span the augmented training pool; the test column is the honest generalization metric for each model.

Model	Test n	CV Acc (mean)	Val Acc	Test Acc	Macro-F1	Wt-F1
MLP (from scratch)	56	0.5846 ± 0.0396	0.4211	0.5000	0.42	0.52
CNN (ResNet18 transfer)	1,255	$0.9998 \pm 0.0003^{\dagger}$	0.6801	0.6661	0.53	0.64

†Inflated by augmented leakage within folds; see Section 4.3.1.

Table 8: Per-class precision, recall, and F1-score on each model's held-out test set. MLP: $n = 56$ (real images only). CNN: $n = 1{,}255$ (real + held-out augmented copies, improved pipeline).

Model	Class	Precision	Recall	F1	Support
MLP ($n = 56$)	negative	0.76	0.54	0.63	35
	moderate	0.29	0.58	0.39	12
	positive	0.29	0.22	0.25	9
CNN ($n = 1{,}255$)	negative	0.75	0.85	0.80	782
	moderate	0.55	0.24	0.33	279
	positive	0.42	0.55	0.48	194

5.1 MLP



Figure 2: Loss per epoch (left) and accuracy per epoch (right) for the MLP model during full-split training (18 epochs before early stopping). Training loss falls steadily toward zero (training accuracy reaching ≈96%), while validation and test loss diverge upward — a clear sign of severe overfitting on the small evaluation sets (57 and 56 samples). Validation and test accuracy remain volatile, fluctuating between roughly 37% and 58%, with a best validation accuracy of 56.1%.



Figure 3: Learning curve showing cross-entropy loss for training, validation, and test sets as the number of training samples grows from 100 to 1,000. Training error falls consistently with more data (from ≈0.99 to ≈0.33), but validation and test errors stay high and erratic (validation rising from ≈1.08 to ≈1.94). This pattern suggests that additional data alone would not close the generalization gap for a flat pixel-based MLP, and points toward the need for convolutional architectures.

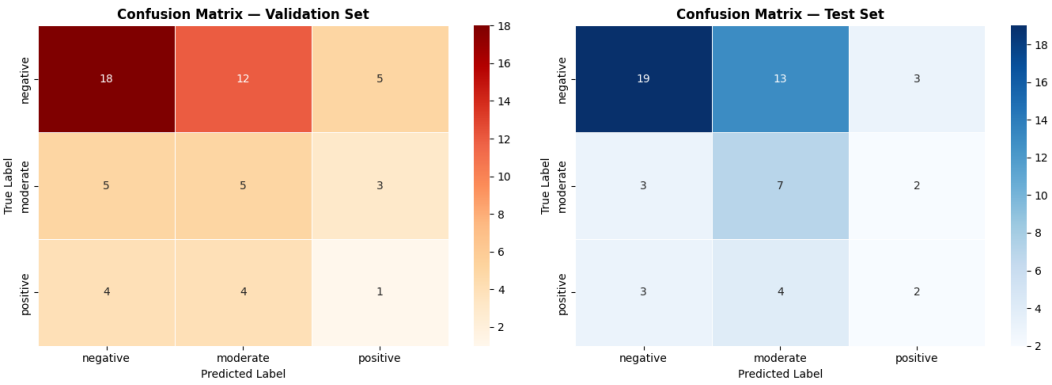


Figure 4: Confusion matrices for the MLP model on the validation set (left, $n = 57$) and the held-out test set (right, $n = 56$). On the test set, negative (compliant) samples are classified most reliably (19 of 35 correct, recall 0.54), the moderate class is over-predicted (7 of 12 correct, recall 0.58, but precision only 0.29), and the positive (violation) class is rarely recovered (2 of 9 correct, recall 0.22), consistent with the class imbalance and the model's reliance on majority-class pixel statistics.

5.2 CNN

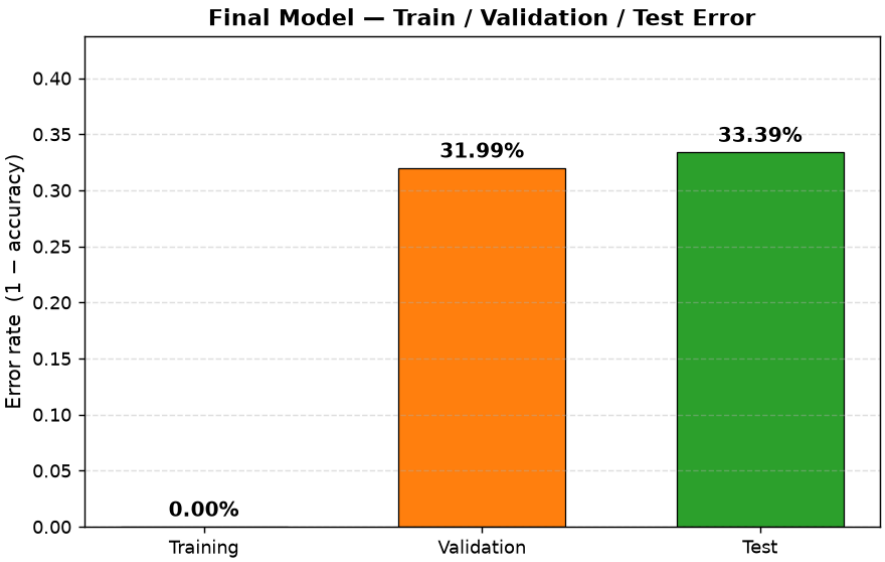


Figure 5: Final train / validation / test error rates (1 − accuracy) for the fine-tuned CNN, measured with a deterministic (non-augmented) transform on all three splits. Training error is 0.00% (the model fully memorizes the 4,467-image training split), while validation error is 31.99% (619 images) and test error is 33.39% (1,255 images) — the honest measures of generalization to unseen buildings.

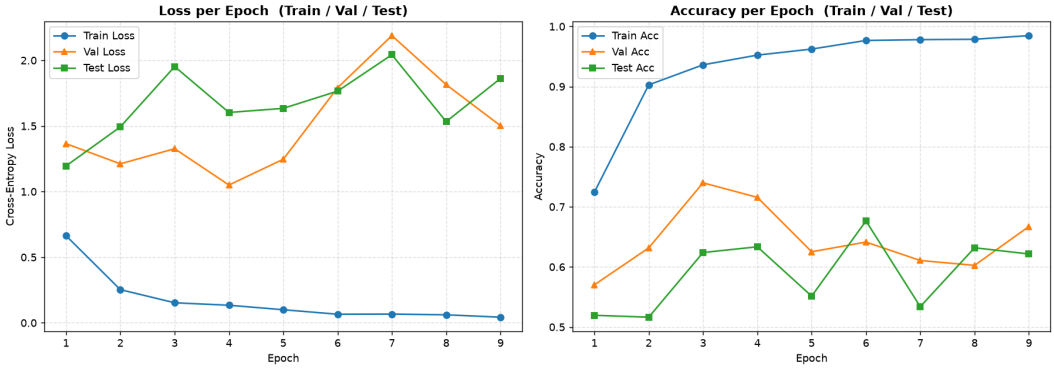


Figure 6: Loss per epoch (left) and accuracy per epoch (right) for a freshly initialized CNN trained directly on the full post-mining training split, evaluated after every epoch on both validation and test sets (9 epochs before early stopping, patience 6). Training loss falls from 0.66 to 0.04 and training accuracy climbs to ≈98%, while validation and test loss remain elevated and volatile (val: 1.05–2.19; test: 1.20–2.04). Validation accuracy peaks at 73.99% (epoch 3) before degrading with continued training; test accuracy tracks a similar noisy pattern, peaking at 67.65% (epoch 6). This instability after the first few epochs is why the final deployed model instead uses the K-fold-selected checkpoint fine-tuned for a short, fixed schedule (Table 5), rather than the longest-trained checkpoint from this run.

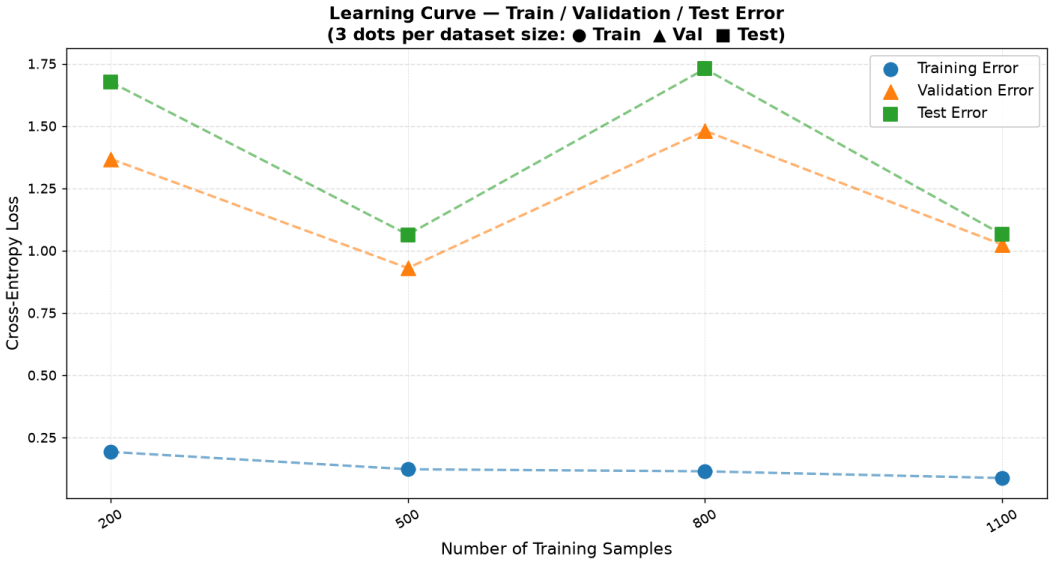


Figure 7: Learning curve showing cross-entropy loss for training, validation, and test sets as the number of training samples grows from 200 to 1,100 (8 epochs per checkpoint). Training loss decreases steadily with more data (0.192 → 0.087), while validation and test loss remain high and non-monotonic (validation: 1.37, 0.93, 1.48, 1.02; test: 1.68, 1.06, 1.73, 1.06), never converging toward the training curve. This indicates that, at these dataset sizes, the generalization gap is dominated by variance rather than by a lack of data alone.

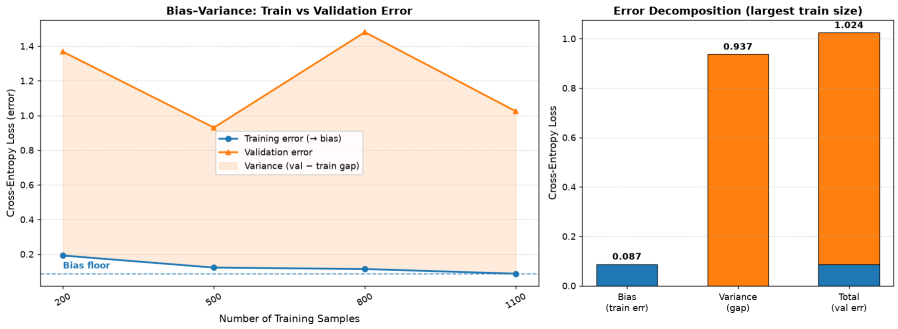


Figure 8: Bias–variance decomposition derived from the learning-curve sweep, using the training-error floor at the largest training size (1,100 samples) as an estimate of bias and the validation-minus-training gap as an estimate of variance. The training-error floor is 0.087, the variance term is 0.937, and the resulting validation error is 1.024 (in cross-entropy loss). The very small bias relative to the large variance term confirms that the CNN's error is dominated by overfitting/generalization variance rather than by insufficient model capacity.

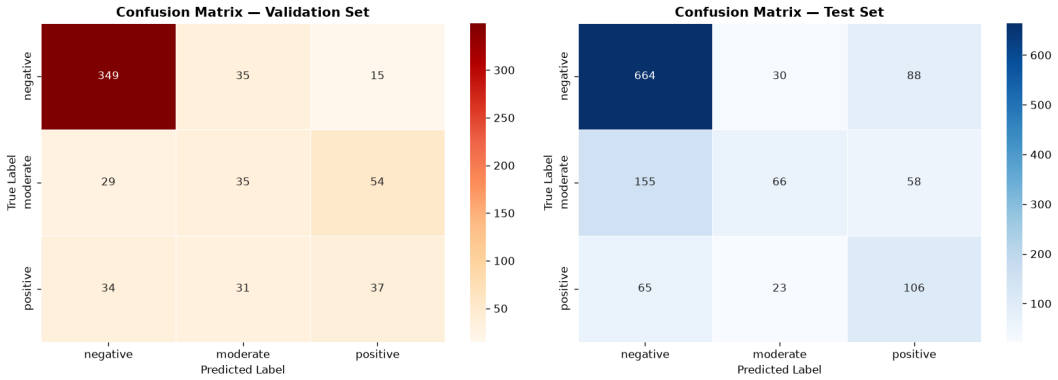


Figure 9: Confusion matrices for the fine-tuned CNN on the validation set (left, $n = 619$) and the held-out test set (right, $n = 1{,}255$). On the test set, the negative (compliant) class is classified most reliably (664 of 782 correct, recall 0.85), the positive (violation) class is recovered with precision 0.42 and recall 0.55 (106 of 194 correct), and the moderate (borderline) class remains the hardest (66 of 279 correct, recall 0.24), with most of its errors going to the negative class (155 of 279). The validation confusion matrix shows a similar pattern at smaller scale (399 negative, 118 moderate, 102 positive samples), confirming that the moderate class is the model's main source of error at both stages.

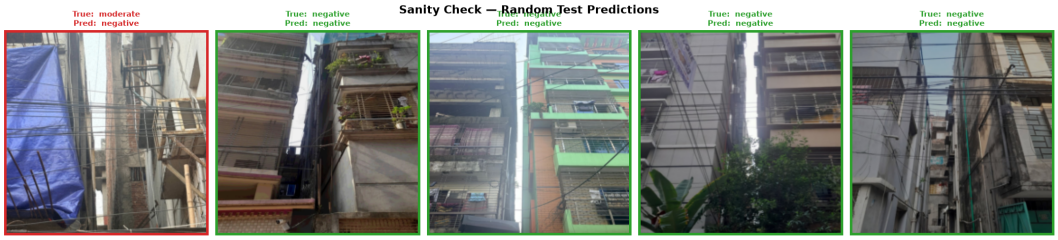


Figure 10: Sanity check: predictions on five randomly sampled test images. Four of five negative (compliant) images are correctly classified; one moderate (borderline) image is misclassified as negative, illustrating the persistent confusion between the moderate and negative classes discussed above.

6 Conclusion

This paper set out to explore whether building setback conditions in Dhaka — a problem that has long resisted effective enforcement — could be identified automatically from street-level photographs. We introduced the RAJUKSetbackBD dataset: 372 smartphone images collected across sixteen neighborhoods, each annotated under a three-class protocol (Positive/Violation, Moderate/Borderline, Negative/Compliant) grounded in RAJUK's setback regulations [2] and the BNBC [3], and each carrying a measured building-to-boundary distance. Using leakage-free, subject-level splitting and weighted cross-entropy

loss, we trained and compared two models: a from-scratch Multi-Layer Perceptron baseline and a ResNet18 transfer-learning model, the latter further refined with a larger 70:20:10 evaluation split and hard-example mining of previously misclassified validation images.

The convolutional model clearly outperformed the flat-pixel baseline, improving held-out test accuracy from 50.0% to 66.6% and macro-F1 from 0.42 to 0.53, with a training/validation/test error profile of 0.00% / 31.99% / 33.39%. The gap between the two confirms that spatial feature extraction — the kind provided by convolutional architectures like ResNet [14], MobileNetV2 [15], and EfficientNet [16] — is genuinely needed for this task. At the same time, the CNN's zero training error alongside a 32–33% validation/test error confirms that overfitting/variance, rather than insufficient model capacity (bias), remains the dominant source of error, and the moderate (borderline) class remained difficult throughout, underscoring that more labeled real images, stronger regularization, and continued hard-example mining are the natural next steps.

Beyond the models themselves, the dataset, annotation protocol, and evaluation methodology established here provide a reproducible foundation for future experiments with deeper convolutional and transfer-learning approaches. Crucially, the data collection method — smartphone photographs taken by field personnel during ordinary surveys — requires no specialized equipment and fits naturally into RAJUK's existing inspection workflows. We hope this makes the approach not only technically interesting, but practically useful. We release the dataset openly [19] in the hope that it will support further research into automated urban compliance monitoring across Dhaka and other rapidly growing South Asian cities facing similar challenges.

References

[1] World Bank. Dhaka: Improving Living Conditions in the Urban Slums. World Bank Urban Development Report, 2023. Available: https://www.worldbank.org/en/country/bangladesh/overview

[2] RAJUK. Dhaka Metropolitan Building Construction Rules (DMBCR). Rajdhani Unnayan Kartripakkha, Dhaka, Bangladesh, 2008. Available: https://rajuk.gov.bd

[3] Ministry of Housing and Public Works. Bangladesh National Building Code (BNBC). Government of Bangladesh, 2020. Available: http://www.mohpw.gov.bd

[4] A. Lubaina, Md. E. U. Pahlowan, T. I. Munni, and M. Ibrahim. FootpathVision: A comprehensive image dataset and deep learning baselines for footpath encroachment detection. *SSRN Preprint*, 2025. Available: https://ssrn.com/abstract=5577201

[5] R. Hossen, D. Mistry, M. Rahman, W. A. S. A. R. Hridoy, S. Saha, and M. Ibrahim. Road damage and manhole detection using deep learning for smart cities: A polygonal annotation approach. *arXiv preprint* arXiv:2510.03797, 2025. Available: https://arxiv.org/abs/2510.03797

[6] Md. H. Ferdaus, R. H. Prito, M. Ahmed, M. M. Islam, Md. S. Ali, M. Ibrahim, A. A. S. Rasel, M. Islam, T. Jabid, and Md. A. Rahman. MangoImageBD: An extensive mango image dataset for identification and classification of various mango varieties in Bangladesh. *Data in Brief*, page 111908, 2025.

[7] Md. Tahsin, M. Ibrahim, A. T. Nafisa, M. B. R. Nuha, M. I. Arnab, Md. H. Ferdaus, M. M. Islam, M. R. A. Rashid, T. Jabid, and Md. S. Ali. PaddyVarietyBD: Classifying paddy variations of Bangladesh with a novel image dataset. *Data in Brief*, 60:111514, 2025.

[8] M. Hasan, M. Ibrahim, and S. Ali. Selecting update blocks of convolutional neural networks using genetic algorithm in transfer learning. *Machine Graphics and Vision*, 33, 2024.

[9] L. Huang, X. Li, and B. Chen. Automatic detection of illegal buildings using high-resolution satellite images and deep learning. *Remote Sensing*, 13(11):2182, 2021. Available: https://doi.org/10.3390/rs13112182

[10] A. Varghese, J. Gubbi, A. Ramaswamy, and P. Balamuralidhar. ChangeNet: A deep learning architecture for visual change detection. In *Proceedings of the ECCV Workshops*, 2018.

[11] H. Maeda, Y. Sekimoto, T. Seto, T. Kashiyama, and H. Omata. Road damage detection and classification using deep neural networks with smartphone images. *Computer-Aided Civil and Infrastructure Engineering*, 33(12):1127–1141, 2018. Available: https://doi.org/10.1111/mice.12387

[12] G. Weld, E. Jang, A. Li, A. Zeng, K. Heimerl, and J. E. Froehlich. Deep learning for automatically detecting sidewalk accessibility problems using streetscape imagery. In *Proceedings of the 21st ACM SIGACCESS*

Conference on ASSETS'19, pages 196–209, 2019. Available: https://doi.org/10.1145/3308561.3353799

[13] M. Hosseini, M. Saugstad, F. Miranda, A. Sevtsuk, C. T. Silva, and J. E. Froehlich. Towards global-scale crowd+AI techniques to map and assess sidewalks for people with disabilities. In *Proceedings of the IEEE/CVF CVPR*, 2022. Available: https://doi.org/10.1109/CVPRW56347.2022.00106

[14] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE CVPR*, pages 770–778, 2016. Available: https://doi.org/10.1109/CVPR.2016.90

[15] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE CVPR*, pages 4510–4520, 2018. Available: https://doi.org/10.1109/CVPR.2018.00474

[16] M. Tan and Q. V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th ICML*, pages 6105–6114, 2019. Available: https://arxiv.org/abs/1905.11946

[17] K. Simonyan and A. Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint* arXiv:1409.1556, 2014. Available: https://arxiv.org/abs/1409.1556

[18] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger. Densely connected convolutional networks. In *Proceedings of the IEEE CVPR*, pages 4700–4708, 2017. Available: https://doi.org/10.1109/CVPR.2017.243

[19] F. Tasnim, F. Yasmin, Md. T.-U.-Rahman, A. Islam, T. B. Tariq, and Y. Bhuiyan. Building Setback Violation Image Dataset, RAJUK (RAJUKSetbackBD). Google Sheets, 2026. Available: https://docs.google.com/spreadsheets/d/1_QZuP0QnBCnsbc-FAWuSNSJL57rJ3GxVhxL3k6gdM8c